

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination- 2020
Course No: 401 (Microeconomic Theory II)
Time: 4 Hours Full Marks: 70

Answer any five questions taking at least one from each part. (5x14=70).
All parts of a question must be answered consecutively

Part A

1. a) 'Pareto optimality is a necessary but not sufficient condition for social welfare maximization' – justify the statement considering a $2 \times 2 \times 2$ model. [7]
- b) Explain the underlying assumptions about the marginal utility of money for being the Hicks – Kaldor compensation be correct? [7]
- ✓ 2. a) In a 2-person 2-good case (assuming the goods have fixed supply), show with appropriate equations how Pareto optimality for consumption will be achieved. [7]
- b) Assume that firm 1 experiences external economies and firm 2 experiences external diseconomies. Both the firms sell their product in a competitive market where, $p_1 = p_2 = 15$. The cost functions of the two firms are as following: $C_1 = 0.1 q_1^2 + 5 q_1 - 0.1 q_2^2$ and $C_2 = 0.2 q_2^2 + 7 q_2 + 0.025 q_1^2$. Show how Pareto optimality can be achieved through the imposition of appropriate taxes and subsidies. Find the social dividend that can be used to increase the utility. [7]
- ✓ 3. a) State and explain the First and the Second Welfare Theorem with their implications. [8]
- b) What is the social welfare function? Explain the difference between Utilitarian welfare function and Rawlsian welfare function. [6]

Part B

4. a) Define Nash equilibrium. Suppose that identical duopoly firms have constant marginal costs of \$10 per unit. Firm 1 faces a demand function of $q_1 = 100 - 2p_1 + p_2$ and Firm 2 faces a demand function of $q_2 = 100 - 2p_2 + p_1$, where p_1, q_1 is the price and quantity of output for Firm 1 respectively. Similarly p_2, q_2 is the respective price and quantity for Firm 2. Solve for the Bertrand equilibrium. How could you explain this as a Nash equilibrium? [8]
- b) What is Prisoners' dilemma? Explain the implications of Prisoners' dilemma for the oligopolistic market? What will be the strategic interaction? [6]
- ✓ 5. a) Say, there are two firms (A and B) that are considering whether or not to enter a new market. However, the market is only big enough to support one of the two firms. If both firms enter the market, then they will both make a loss of £ 20m. If only one firm enters the market, that firm will earn a profit of £50m., and the other firm will just break even. We also assume that firm B observes whether firm A has entered the market before it decides what to do. From this above information, present the game in extensive form and applying the principle of backward induction show that the outcome of the game will be – Firm A will enter the market, and Firm B will stay out. [6]

b) From the following game, show that the solution "down/right" can be a subgame perfect Nash Equilibrium if the game is infinitely repeated and both players adopt the following strategy- play right-down initially or if this has always been played in the past. Otherwise, play Up/left.

		Player 2	
		Left	Right
Player 1	Up	4,4	6,3
	Down	3,6	5,5

[8]

6. a) What do you understand by the Paradox of backward induction? Explain how the existence of multiple Nash equilibrium can help to avoid this paradox. [8]

b) If players initially co-operate and play across the paradox of backward induction is partially overcome -explain this statement with the help of Centipede game developed by Rosenthal. [6]

Part C

✓ 7. a) What do we mean by Tragedy of Commons? Explain with an example. [3]

b) Say, there are two firms- A and B. Firm A is in the upstream of a river that produces steel and some amount of pollution, x . Whereas, firm B, a fishery, is in the downstream and is adversely affected by the pollution. Increase in pollution increases the cost of firm B, however, it reduces the cost of firm A. From this above information show the private and social costs and the production level under the following situations- if firms maximize profit- i) individually, ii) jointly, and iii) under different property rights. [11]

✓ 8. a) Show that asymmetric information may lead to disappearance of quality of good from the market. [4]

b) By using Michael Spence's Model of Education explain how the level of education can work as a signal of the different productivity of the workers. [10]

Department of Economics
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Course No: 402 (Macroeconomic Theory-II)
Time: 4 Hours Full Marks: 70

Answer any five (05) of the following questions. All parts of a question must be answered consecutively

1. a) What were the Kuznets' findings on aggregate consumption and saving behavior in the 1950s? How did these findings contradict the Keynesian theory of consumption? Discuss. [4]
- b) Illustrate the intertemporal optimizing model of consumption developed by Irving Fisher. Establish how the ratio of marginal utilities of consumption in every two adjacent periods over time is equal to the ratio of the market interest rate to the consumer discount rate. Display its implications by applying an alternative time profile of consumption. [10]
2. a) Discuss the permanent income hypothesis (PIH) developed by Milton Friedman. How is his theory different from that proposed by J.M. Keynes? How does Friedman's theory of consumption resolve the apparent consumption between LR (Long Run) and SR (Short Run) observations, as confirmed by Kuznets' findings? [10]
- b) What is the "Ratchet Effect" of consumption? Describe the essential elements of the Relative Income Hypothesis (RIH). [4]
3. a) Explain the determinants of investment using equations. [4]
- b) Derive the investment function. [4]
- c) Prove that in the long run the marginal product of capital equals the real cost of capital. [4]
- d) Define Tobin's q . [2]
4. a) How did Tobin differentiate among individual preferences in terms of portfolio decisions? Describe using diagram(s) how the change in the expected capital gains and risks changes the aggregate demand for money in this approach? [8]
- b) Derive the optimal number of money transactions developed by Baumol-Tobin in their transaction demand approach. [6]
5. a) Explain in detail the meaning of high powered money. Why is it called so? Use equations and examples. [6]
- b) How does the Federal Reserve System make the economy more stable by aiming for a particular money stock or for a particular interest rate? [6]
- c) Show the equilibrium in the money market and the effects of a decrease in the high powered money. [2]

6. a) How well will Fiscal and Money rules achieve internal and external balance under a fixed or flexible exchange rate? Explain. [5]
b) Should fiscal or monetary policy be targeted at internal or external equilibrium? Examine using the fixed exchange rate system. [7]
c) What is the leakage effect? [2]
7. a) "On the balanced growth path, the growth rate of output per worker is determined solely by the rate of technological progress"- Clarify using the Solow growth model. [4]
b) Demonstrate the impacts of saving ratio in the steady-state solution and prove that anything other than changes in the technology leaves only level effects on k and y , but not the growth effects. [5]
c) Exhibit the effects of changes in the saving ratio on per capita consumption. Does any steady-state solution automatically imply the golden rule of capital accumulation? Discuss. [5]

✓ 8. Write short notes on any four (04) of the following:

[4×3.5=14]

- a) PV criteria for investment
- b) Instrument problem
- c) Life-Cycle Hypothesis
- d) Regressive expectation model
- e) Current and capital account of BOP
- f) Inventory Investment

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Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination-2020
Course No: 403 (Econometrics II)
Time: 4 Hours **Full Marks: 70**

[Answer any FIVE (05) of the following questions. All parts of a particular question MUST be answered consecutively]

- [4+4+3+3]
- ✓ 1. Assume that you have estimated a demand for roses function for the Dhaka city over the period 2015 Q1 – 2022 Q1. The regression model has been in log-linear form, where the use of OLS method has produced the following estimates:

$$\begin{aligned}\widehat{\ln Y_t} &= 9.23 - 1.76 \ln X_{2t} + 1.34 \ln X_{3t} \\ SE &= (\dots) \quad (0.30) \quad (\dots) \\ t &= (16.23) \quad (\dots) \quad (2.54) \\ p \text{ value} &= (0.000) \quad (0.000) \quad (0.003) \\ R^2 &= 0.729 ; \quad Adj R^2 = 0.715 ; \quad df = 26 ; \quad F = 17.50\end{aligned}$$

where, Y is the quantity of roses in dozens, X_2 is the average wholesale price of roses (TK/dozen), and X_3 is the average wholesale price of tuberose (TK/dozen).

- i) Fill in the blanks for values and provide appropriate interpretation of the estimates.
- ii) Validate whether the parameters of the model are individually significant or not. Use both 't' and 'p' values to justify your arguments.
- iii) "Imposing restriction in the model reduces the goodness of the fit" – do you agree with this from the above results?
- iv) Highlight the situations when you cannot compare the results of two regression models based on R^2 measures.

- [6+8]
- ✓ 2. a) How would you test the overall significance of a model? From the above regression results, validate that the slope parameters of the model are jointly significant.

b) Consider a savings-income relationship for Bangladesh over the period 1970-1995. Given that the country adopted a massive 'structural adjustment' policy for its economy in 1982, how would you describe the 'Chow test' procedure for testing the structural stability of the regression model depicting the aforesaid relationship?

[4+3+2+2+3]

3. a) What is a binary or dummy variable and what is the dummy variable trap? Explain.
- b) Using the following estimated model answer the following questions:

$$\begin{aligned}\log(\widehat{wage}) &= 0.56 - 0.324 \text{ female} + 0.087 \text{ educ} - 0.0067 \text{ female.educ} + 0.034 \text{ exper} - 0.00063 \text{ exper}^2 \\ &\quad (0.32) \quad (0.16) \quad (0.036) \quad (0.0021) \quad (0.013) \quad (0.00033) \\ n &= 526 \quad R^2 = 0.563\end{aligned}$$

Where, wage=\$/hour educ, exper = education and experience in years

- I. Define the control group and experimental or treatment group. Why they are?
- II. All other factors being equal, is there evidence that male can earn more than female? How strong is the evidence?
- III. What is the return of female when her educ is 10 years? Interpret it.
- IV. At what point does the marginal effect of wage on experience become negative?

[7+7]

4. a) Consider the results from a regression of household consumption expenditure on their income and wealth:

$$\hat{Y}_i = 24.7747 + 0.9415 X_{2i} - 0.0424 X_{3i}$$

$$SE = (6.7525) \quad (0.8229) \quad (0.0807)$$

$$t = (3.6690) \quad (\dots\dots) \quad (\dots\dots)$$

$$R^2 = 0.9635 \quad \text{Adj } R^2 = 0.9531 \quad df = 7$$

Filling in blank spaces in the brackets, state clearly with reasons, whether this model suffers from multicollinearity problem or not. If yes, how could you come out of this?

- b) Highlighting that OLS estimators remain BLUE in the presence of multicollinearity, illustrate how large variances/covariances and wider confidence intervals make estimation and hypothesis testing difficult.

[2+7+5]

5. a) What is the generalised least squares (GLS) method?

- b) Explain how you can obtain GLS estimators for the 2-variable regression model:

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

What difference do you find between OLS and GLS/ WLS estimators?

- c) Describe how you can detect the presence of heteroscedasticity in the regression model, following the Goldfeld-Quandt testing procedure.

[4+5+5]

6. a) Mention the reasons for autocorrelation to occur. In particular, briefly discuss the cases of 'cobweb phenomenon' and 'autoregression'.

- b) Given the Durbin-Watson d-statistic to test the presence of autocorrelation, explain how you would arrive at the bounds of d [0, 4]? What decision rules would you follow?

- c) Assuming the error term to follow AR (1) scheme, explain how you would remedy 'serial correlation', when the coefficient of autocorrelation is known.

[4+6+4]

7. a) What are the consequences of including an irrelevant variable in a regression model?

- b) What procedures would you follow for using Ramsey's RESET test to come to a decision about specification error in a regression model?

- c) What advantages do you find while utilizing AIC and SIC criteria, over simple R^2 criterion, for selecting competing models?

8. Explain the following statements (any FOUR):

[3.5X4 = 14]

- ✓ a) Beta coefficient
- ✓ b) Transformation of variables may reduce multicollinearity in the regression model.
- c) Breusch-Godfrey (BG) test for autocorrelation.
- ✓ d) There are ways to come out of the dummy variable trap.
- e) Slope coefficient(s) in a log-linear model shows elasticity.
- ✓ f) ANCOVA model.

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination 2020
Course No: 404 (Health Economics)
Time: 4 Hours Full Marks: 70

Answer any FIVE of the following questions. All parts of a particular question must be answered consecutively

1. a) Define health economics. Does economics apply to health and health care? [6]
b) Why is medical care different from other goods and services? Explain. [8]
2. a) If not medicine, what factors were responsible for the historical decline in the population mortality rate? Explain. [8]
b) Explain the two different theories of schooling. [6]
3. a) Define production function. Describe the production function of health. [6]
b) Would health production function eventually bend downward? Explain in support of your answer. [8]
4. a) Explain the potential use of a person's time by labor-leisure trade off. [4]
b) Is it possible to earn more money and more leisure time together? Explain in the light of Grossman model. [10]
5. a) Explain the determinants of demand for health care visit. [8]
b) What is the role of time and coinsurance on the health care demand? [6]
6. a) Distinguish between CEA and CBA. [3]
b) Explain (with a hypothetical example) how you would derive the decision rule for an optimal resource allocation in health interventions when programs are independent or mutually compatible. [3]
c) What is the complication that arises while identifying the optimal resource allocation for these types of health interventions? [8]
7. a) How do asymmetric information arise inefficiency problems in the health sector? [3]
b) Explain asymmetric information using Lemons Principle. [5]
c) Show the application of Lemons Principle in health insurance market. [6]
8. Write short notes on any two of the following: [7x2=14]
 - a) Cost Utility Analysis (CUA)
 - b) Supplier Induced Demand (SID)
 - c) Cost Effectiveness Analysis (CEA)
 - d) Role of age and wage on optimal health stock.

Department of Economics
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4th Year (Hons.) Final Examination 2020
Course No.: 405 (Development Experience and Methods)
Time: 4 Hours Full Marks: 70

Answer any five of the following questions. All parts of a particular question must be answered consecutively.

1. a) What do you mean by "Mode of Production"? Describe the role of mode of production to explain the transition process of an economy. [6+8]
b) Define Capitalism and Socialism. Explain the transition process from Capitalism to Socialism for a country.
2. a) What are the different meanings of Transition in Development perspective? [4+8+2]
b) What are the internal & external conditions for the industrial revolution in United Kingdom? Briefly explain.
c) Is there any social cost of industrial revolution in UK? Explain your answer.
3. What are the causes of Great Depression in USA? Do you think New Deal program played an effective role to recover the depressed situation? Why or why not? [4+6+4] 14
4. a) What do you mean by market socialism? Explain the organization of market socialism in Lange Framework. [4+3+7]
b) Describe the Great Leap Forward as first upheaval in China.
c) What do you mean by Middle Income Trap? Will Inequality lead China to the Middle-Income Trap?
5. a) Define Translative Adaptation. Describe the role of translative adaptation for the success as developed one through the historical path traveled by Japan as a developing country. [8+6]
b) What do you mean by War Economy? What are the reasons for the transition of Japan as a post-war economic miracle? Briefly explain.
6. a) What do you mean by welfare state? Do you think that Sweden is one of examples of welfare state? Why? Is there any barrier as policy option for being a welfare state? Explain. [8+6]
b) Briefly describe the Education system in Sweden.
7. a) What do you mean by East Asian Miracle? [4+10]
b) Define export-oriented industrialization and import oriented industrialization? Describe the development experience of South Korea and Taiwan. What are similarities and differences as transition mechanism between these two countries?
8. a) Define center and periphery as development perspective. Briefly explain the transition of Cuba from peripheral economy. How could revolution play role for the transition in an economy? Give your answer regarding Cuba. [8+6]
b) Briefly compare the health system of Sweden and Cuba.

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination-2020
Course no: 406 (Money, Financial Institutions and Capital Market)
Time: 4 Hours Full marks: 70

Answer any five (05) of the following questions
All parts of a particular question must be answered consecutively

1. Briefly explain the following terms: [14]
a) Captive finance (b) Closed-end fund (c) Debt deflation (d) Disintermediation (e) Exercise price
(f) FICO scores (g) Hybrid funds (h) Liquidity premium theory (i) Load fund (j) Securitized mortgage (k) Segmentation theory (l) Notional principal (m) Private equity buyout (n) Restrictive covenants.
2. ✓ a) When was the Financial Sector Reform Program (FSRP) taken? Describe the objectives and measures of FSRP. [4]
b) How would you define NBFIs in Bangladesh perspective? What are the indicators of CAMELS rating? How does CAMELS rating evaluate the performance of NBFIs in Bangladesh? Explain. [5]
c) What are the instruments used by Bangladesh Bank to implement monetary policy in Bangladesh? Briefly explain role of CRR and SLR in controlling the money supply in Bangladesh. [5]
3. ✓ a) What is law of one price? How may you derive the theory of Purchasing Power Parity (PPP) from the law of one price? Why does the PPP theory fail to explain exchange rates fully? [4]
b) What are the determinants of short-run equilibrium exchange rate? How do the determinants affect the exchange rate? [5]
c) Explain the factors that affect equilibrium exchange rate in the long run. [5]
4. ✓ a) What are the characteristics of financial assets? Explain. [5]
b) Explain different ways in which various types of financial transactions occur in the economy. [5]
c) Discuss the various types of functions of financial intermediaries. [4]
5. ✓ a) Illustrate the characteristics of money market. Who are the major participants in the money market? Discuss their roles briefly. [6]
b) What is meant by T-account? Briefly discuss the components of a balance-sheet of Central Bank. [4]
c) Briefly discuss the various functions performed by the Bangladesh Bank. [4]
6. a) Portray the structure of the capital market in Bangladesh. Discuss the category-wise characteristics of the enlisted companies in the capital market of Bangladesh. [6]
b) Diagnose the problems recently created in the stock market of Bangladesh. What measures should be taken to resolve the reported problems. Critically discuss. [8]
7. a) Define money laundering as per Money Laundering Prevention Act, 2012. How does a money launderer whiten the black money through the banking channel? Explain. [7]
b) The criminal offences like terrorist financing and money laundering are expanding at an alarming rate all over the world. How would you suggest for combating against such types of criminal activities in the context of Bangladesh? Critically explain. [7]
8. ✓ a) Why banking reforms were necessary following independence of Bangladesh in 1971? Explain. [6]
What are the banking reform regimes in Bangladesh? Under which reform program government did denationalize State-owned Commercial Banks (SCBs)?
b) What features of a developing economy financial market make it more difficult for the country's central bank to protect their currency? [5]
c) What are the bases of loan classification in Bangladesh? Briefly discuss. [3]

Department of Economics
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4th Year (Hons.) Final Examination 2020
Course No.: 407 (International Economics)
Time: 4 Hours Full Marks: 70

Answer any FIVE of the following questions. All parts of a particular question must be answered consecutively.

- ✓ 1. a) Briefly explain the components of Balance of Payments. 4
- b) Suppose you are a citizen of Bangladesh. Now consider the following transactions: 10
- ✓ i. Sales overseas of Music CDs valued at Tk.15 million.
 - ✓ ii. Tuition fee of Tk.30 million received by US universities from Bangladeshi students.
 - ✓ iii. Payment of Tk.8 million consulting fees to BIDS by a Mexican firm.
 - ✓ iv. Purchase of Tk.32 million worth British Pounds by the Bangladesh Bank.
 - ✓ v. Purchase of Tk.6 million of British government bonds by the Bangladesh Bank.
 - ✓ vi. Sale of a Tk.100 million Eurobond issued in London by a Bangladeshi firm.
 - ✓ vii. Payment of Tk.10 million in interest by Beximco to its bond investors in London.
 - ✓ viii. Sale of Tk.8 million in gold by the Bangladesh Bank.
 - ✓ ix. Purchase of Tk.25 million in East European stocks by Bangladesh Bank.

Now find out the value of Balance of Trade, Current Account Balance, Capital account Balance and Official Reserve Balance.

- ✓ 2. a) Assume the following exchange rates: 4
- \$2 = £1 in New York
¥410 = £1 in London
¥200 = \$1 in Tokyo
- You have \$50,000 in your account. Indicate how profitable triangular, or three-point, arbitrage can take place?
- b) Calculate the Forward Discount or Forward Premium on yearly basis for the following spot and three-month forward rates: 3
- i. SR = \$2.00/£1 and FR = \$2.01/£1
 - ii. SR = \$2.00/£1 and FR = \$1.96/£1
- c) What do you mean by covered interest arbitrage (CIA)? Describe CIAP (Covered Interest Arbitrage Parity) with graph and CIAM (Covered Interest Arbitrage Margin) with appropriate example. 7
3. a) Explain Asset Market Model. How it differs from Monetary approach? 3
- b) How is exchange rate determined under Portfolio Balance approach? 4
- c) What is interest parity condition? How is equilibrium established in domestic money market and foreign exchange market simultaneously? How will you address the effect of sudden increase in dollar price in money market and foreign exchange market in recent days? 7
- ✓ 4. a) Derive DD-AA schedule. Explain the factors that can cause to shift DD schedule. 5
- b) If Bangladesh government takes fiscal or monetary policy permanently then how the economy will react? 5
- c) Explain Exchange rate overshooting. 4

5. a) Derive Aggregate Demand curve for an open economy under fixed and flexible exchange rate. How it differs from the Aggregate Demand for a closed economy? 6
- b) Assume Bangladesh is experiencing a reduction in capital outflow due to higher interest rate than abroad. How does it affect BD's aggregate Demand under Fixed and Flexible exchange rate? 8
6. a) What factors are important for an individual to set his demand for money? How equilibrium interest is determined? 6
- b) Differentiate between the effects of permanent increase in money supply in the short run and long run. 8
7. a) What is the Law of One Price? What is the relationship between Purchasing Power Parity (PPP) and Law of One Price? 4
- b) Illustrate the monetary approach to the exchange rate model. Why is it considered as a long run model? 7
- c) What predictions do monetary approaches make about the long run effects on the exchange rate of money supplies, interest rates and output level? 3

8. Write short notes: any four

[4x3.5=14]

- a) J-curve effect.
- b) Foreign trade multiplier
- c) Asian Currency in 1997
- d) Marshall-Lerner condition and its relevance today.
- e) Effectiveness of stabilization policy under floating exchange rate.

Tanvir-

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination- 2020
Course No: 408 (Economic Thoughts in Perspective)
Time: 4 Hours Full Marks: 70

Answer any five of the following questions [14x5=70]
All parts of a question must be answered consecutively

১. অর্থশাস্ত্রের বিকাশ ধারার একটি সারসংক্ষেপ লিখ। ইউরোপ কীভাবে অর্থশাস্ত্র চর্চায় মূলধারায় পরিণত হয়? ব্যাখ্যা কর। [১৪]

Write a brief overview of the development of Economics. How did Europe become mainstream in Economics studies? Explain.

- ✓ ২. কোটিল্যের অর্থশাস্ত্র পাঠে প্রাপ্ত জ্ঞান থেকে প্রাচীন ভারতের সমাজ, অর্থনীতি ও রাষ্ট্র বিষয়ে তোমার ধারণা লিখ। ইবনে খলদুনের তৎকালীন অর্থনীতি বিষয়ে বিশ্লেষণ ও প্রধান কয়টি সিদ্ধান্ত লিখ। [১৪]

Elaborate your ideas on society, economy and state of ancient India from your knowledge obtained from Kautilya's Arthashastra studies. Write on Ibne Khaldun's analysis and major conclusions on contemporary economic issues.

- ✓ ৩. শ্রেণী, অদৃশ্য হস্ত এবং অবাধ অর্থনীতি সম্পর্কে এডাম স্মিথের তাত্ত্বিক চিন্তা ব্যাখ্যা কর। তিনি কীভাবে সরকারের দায়িত্ব নির্দেশ করেছিলেন? [১৪]

Elaborate Adam Smith's theoretical ideas on class, invisible hand, and laissez faire. How did he point out the duties of the government?

- 2020-21
৪. ডেভিড রিকার্ডের মূল্যতত্ত্ব বিশ্লেষণ কর। রিকার্ডো ও ম্যালথাসের বিতর্কের বিভিন্ন বিষয় নিয়ে আলোচনা কর। [১৪]

Analyse David Ricardo's theory of value. Discuss issues of debates between Ricardo and Malthus.

৫. মার্কসের অর্থনীতি চিন্তার পেছনে বিভিন্ন বুদ্ধিবৃত্তিক প্রভাব চিহ্নিত কর। মার্কসের তাত্ত্বিক কাঠামো ব্যাখ্যা কর। [১৪]

Identify different intellectual influences on Marxian Economic Thought. Elaborate Marx's theoretical framework.

- ✓ ৬. প্রান্তিকতাবাদীদের প্রধান সিদ্ধান্তসমূহ লিখ। উইলিয়াম থম্পসন কীভাবে 'ব্যক্তি প্রতিযোগিতার বিধ' বর্ণনা করেছেন? আলোচনা কর। [১৪]

Elaborate major articulations of the Marginalists. How did William Thompson describe 'evils of individual competition'? Discuss.

৭. অলফ্রেড মার্শালের তাত্ত্বিক কাঠামো থেকে কীভাবে ব্যক্তি অর্থশাস্ত্রের উদ্ভব হয়েছে এবং কেইনসের তাত্ত্বিক কাঠামো কীভাবে সামগ্রিক অর্থশাস্ত্র তৈরি করেছে? ব্যাখ্যা কর। [১৪]

How did Alfred Marshall's theoretical framework develop Microeconomics and Keynes's that to Macroeconomics? Explain.

- ✓ ৮. সংক্ষেপে ব্যাখ্যা কর (যেকোন দুইটি): (ক) বাণিজ্যতত্ত্ব (খ) বেনথামের উপযোগবাদ (গ) নারীবাদী অর্থশাস্ত্র [৭x২=১৪]

Write short notes (any two): (a) Mercantilist; (b) Bentham's Utilitarianism; (c) Feminist Economics.

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination 2020
Course No.: 409 (Research Methodology and Empirical Studies)
Time: 4 Hours Full Marks: 70

Answer any **FIVE (05)** of the following questions
All parts of a particular question must be answered consecutively

- ✓ 1. a) What is research? Why do we research? [4]
✓ b) Briefly describe the different steps involved in a research process. [10]
- ✓ 2. a) What is sampling? Mention the advantages of sampling. [4]
b) Discuss the various methods of data collection. What are the basic differences between primary and secondary data? [10]
- ✓ 3. a) What are the purposes of literature review? [4]
b) Explain the various types of research? Distinguish between basic and applied research. [10]
- ✓ 4. a) What is Research Design? [4]
b) Explain the importance and significance of a research design? [10]
5. a) Under what circumstances stratified random sampling design is considered appropriate? [6]
b) How would you select a stratified random sample? Explain by means of an example. [8]
6. a) What are the essences of non experimental and experimental research design? [4]
b) Distinguish between experimental and non experimental research design. [10]
- ✓ 7. a) What is research proposal? [4]
✓ b) Briefly explain the techniques of writing research proposal. [10]
8. Write short notes (any four) [3.5x4=14]
 - a) Academic research vs report
 - b) Research problem
 - c) Longitudinal analysis
 - d) Deductive and inductive thinking
 - e) Experimental & Control groups
 - f) Treatments